Gating System

1. Riser is designed so as to
   (a) freeze after the casting freezes
   (b) freeze before the casting freezes
   (c) freeze at the same time as the casting
   (d) minimize the time of pouring

   [ME 1987 : 1 Mark]

2. The contraction allowance provided on the pattern and core boxes compensates for the following type of contraction
   (a) liquid contraction
   (b) solidification contraction
   (c) solid contraction
   (d) All of the above three types of contractions

   [ME 1988 : 1 Mark]

3. Chills are used in moulds to
   (a) achieve directional solidification
   (b) reduce possibility of below holes
   (c) reduce the freezing time
   (d) smoothen the metal for reducing spatter

   [ME 1989 : 1 Mark]

4. Two cubical castings of the same metal and size of 2 cm side and 4 cm side are moulded in green sand. If the smaller casting solidifies in 2 mins, the expected time of solidifications of large casting will be
   (a) 16 min (b) $2\sqrt{2}$ min
   (c) 8 min (d) 4 min

   [ME 1989 : 2 Marks]

5. The pressure at the in-gate will be maximum with the gating system
   (a) 4 : 8 : 3 (b) 1 : 3 : 3
   (c) 1 : 2 : 4 (d) 1 : 2 : 1

   [ME 1990 : 1 Mark]

6. When there is no room temperature change, the total shrinkage allowance on a pattern is independent of
   (a) pouring temperature of the liquid metal
   (b) freezing temperature of the liquid metal
   (c) the component size
   (d) coefficient of thermal contraction of solidified metal

   [ME 1991 : 1 Mark]

7. Converging passage is used for feeding t liquid molten metal into the mould to
   (a) increase the rate of feeding
   (b) quickly break off the protruding portio the casting
   (c) decrease wastage of cast metal
   (d) avoid aspiration of air

   [ME 1991 : 1 Ma

8. Light impurities in the molten metal prevented from reaching the mould cavity providing
   (a) Strainer (b) Bottom well
   (c) Skim bob (d) All of the above

   [ME 1996 : 1 Ma

9. Hardness of green sand mould increases v
   (a) increase in moisture content beyon percent
   (b) increase in permeability
   (c) decrease in permeability
   (d) increases in both moisture content permeability

   [ME 2003 : 1 Ma

10. With a solidification factor of 0.97 × 106 s the solidification time (in seconds) f spherical casting of 200 mm diameter is
    (a) 539 (b) 1078
    (c) 4311 (d) 3233

    [ME 2003 : 2 Ma

11. Gray cast iron blocks 200 x 100 x 10 mm a be cast in sand moulds, Shrinkage allow for pattern making is 1%. The ratio o volume of pattern to that of the casting w
    (a) 0.97 (b) 0.99
    (c) 1.61 (d) 1.03

    [ME 2004 : 2 M

12. A mould has downsprue whose length is and the cross sectional area at the base downsprue is 1 cm^2. The downsprue fe horizontal runner leading into the mould of volume 1000 cm^3. The time required the mould cavity will be
    (a) 4.05 s (b) 5.05 s
    (c) 6.05 s (d) 7.25 s

    [ME 2006 : 2 M

50. In sand casting of hollow part of lead, a cylindrical core of diameter 120 mm and height 180 mm is placed inside the mould cavity. The densities of core material and lead are 1600 kg/m^3 and 11300 kg/m^3 respectively. The net force (in N) that tends to lift the core during pouring of molten metal will be

(a) 19.7 (b) 64.5

(c) 193.7 (d) 257.6

[PI 2008 : 2 Marks]

51. A solid cylinder of diameter D and height equal to D, and a solid cube of side L are being sand cast by using the same material. Assuming there is no superheat in both the cases, the ratio of solidification times of the cylinder to the solidification time of the cube is

(a) $(L/D)^2$ (b) $(2L/D)^2$

(c) $(2D/L)^2$ (d) $(D/L)^2$

[PI 2009 : 2 Marks]

52. Solidification time of a metallic alloy casting is

(a) Directly proportional to its surface area

(b) Inversely proportional to the specific heat of the cast material

(c) Inversely proportional to the thermal diffusivity of the mould material

(d) Inversely proportional to the pouring temp

[PI 2010 : 1 Mark]

53. During the filling process of a given sand mould cavity by molten metal through a horizontal runner of circular C.S, the frictional head loss of the molten metal in the runner will increase with the

(a) Increase in runner diameter

(b) Decrease internal surface roughness of the runner

(c) Decrease in length of runner

(d) Increase in average velocity of molten metal

[PI 2010 : 2 Marks]

54. In sand casting process, a sphere and a cylinder of equal volumes are separately cast from the sand molten metal under identical conditions. The height and diameter of the cylinder are equal. The ratio of the solidification time of the sphere to that of the cylinder is

(a) 1.14 (b) 0.87

(c) 1.31 (d) 0.76

[PI 2011 : 2 Marks]

55. A mould having dimensions 100 mm × 90 mm × 20 mm is filled with molten metal through a gate with height h and C.S. area A, the mould filling time is t_1 The height is now quadrupled and the cross-sectional area is halved. The corresponding filling time is t_2. The ratio t_2/t_1 is

(a) $1\sqrt{2}$ (b) 1

(c) $\sqrt{2}$ (d) 2

[PI 2012 : 2 Marks]

Casting Method

56. Increase in water content in moulding sand causes

(a) flowability to go through a maxima

(b) permeability to go through a maxima

(c) compressive strength to go through a maxima

(d) strength to go through a maxima

[ME 1989 : 1 Mark]

57. In a green sand moulding process, uniform ramming leads to

(a) less chance of gas porosity

(b) uniform flow of molten into the mould cavity

(c) greater dimensional stability of the casting

(d) less sane expansion type of casting defect

[ME 1992 : 1 Mark]

58. Casting process

A. Slush casting

B. Shell moulding

C. Dry sand moulding

D. Centrifugal casting

Product

1. Turbine blade

2. Machine tool bed

3. Cylindrical block

4. Hollow castings like lamp shades

5. Rain water pipes

6. Cast iron shoe brake

[ME 1992 : 2 Mark

59. Centrifugally casted products have

(a) large grain structure with high porosity

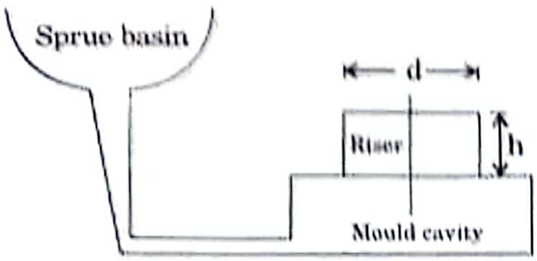
(b) fine grain structure with high density

(c) fine grain structure with low density

(d) segregation of slug toward the outer sk of the casting

[ME 1993 : 1 Ma

20. A cylindrical blind riser with diameter d and height h, is placed on the top of the mold cavity of a closed type sand mould as shown in the figure. If the riser is of constant volume, then the rate of solidification in the riser is the least when the ratio h : d is



(a) 1:2 (b) 2:1
(c) 1:4 (d) 4:1

[ME 2014 : 2 Marks, Set-3]

21. A cylindrical riser of 6 cm diameter and 6 cm height has to be designed for a sand casting mould for producing a steel rectangular plate casting of 7 cm × 10 cm × 2 cm dimensions having the total solidification time of 1.36 minute. The total solidification time (in minute) of the riser is _____.

[ME 2014 : 2 Marks, Set-4]

22. The solidification time of a casting is proportional to $\left(\frac{V}{A}\right)^2$ where V is the volume of the casting and A is the total casting surface area losing heat. Two cubes of same material and size are cast using sand casting process. The top face of one of the cubes is completely insulated. The ratio of the solidification time for the cube with top face insulated to that of the other cube is

(a) $\frac{25}{36}$ (b) $\frac{36}{25}$

(c) 1 (d) $\frac{6}{5}$

[ME 2015 : 2 Marks, Set-1]

23. A cube and a sphere made of cast iron (each of volume 1000 cm^3) were cast under identical conditions. The time taken for solidifying the cube was 4 s. The solidification time (in s) for the for the sphere is _______.

[ME 2015 : 2 Marks, Set-2]

24. Ratio of solidification time of a cylind casting (height = radius) of that of a c casting of side two times the heigh cylindrical casting is_______

[ME 2015 : 2 Marks, S

25. The dimensions of a cylindrical side (height = diameter) for a 25 cm × 15 cm × steel casting are to be determined. Fo tabulated shape factor values given b diameter of the riser (in cm) is_______.

Shape Factor	2	4	6	8	10
Riser volume/ Casting volume	1.0	0.70	0.55	0.50	0.40

[ME 2015 : 2 Marks, S

26. The part of a gating system which reg the rate of pouring of molten metal is

(a) pouring basin (b) runner
(c) choke (d) ingate

[ME 2016 : 1 Mark, S

27. A cylindrical job with diameter of 200 m height of 100 mm is to be cast using mo method of riser design. Assume that the b surface of cylindrical riser does not cont as cooling surface. If the diameter of the is equal to its height, then the height riser (in mm) is

(a) 150 (b) 200
(c) 100 (d) 125

[ME 2016 : 2 Marks,

28. Equal amounts of a liquid metal at the temperature are poured into three m made of steel, copper and aluminum. The of the cavity is a cylinder with 15 mm dia The size of the moulds are such that the temperature of the moulds do not in appreciably beyond the atmos temperature during solidification sequence of solidification in the moul the fastest to slowest is (Thermal conduc of steel, copper and aluminum are 60 and 237 W/mK, respectively. Specific steel, copper and aluminum are 434, 903 J/ kgK, respectively. Densities c copper and aluminum are 7854, 8933 a kg/m^3, respectively.)

(a) Copper - Steel - Aluminium
(b) Aluminum - Steel - Copper
(c) Copper - Aluminium - Steel
(d) Steel-Copper-Aluminum

[ME 2016 : 1 Mark

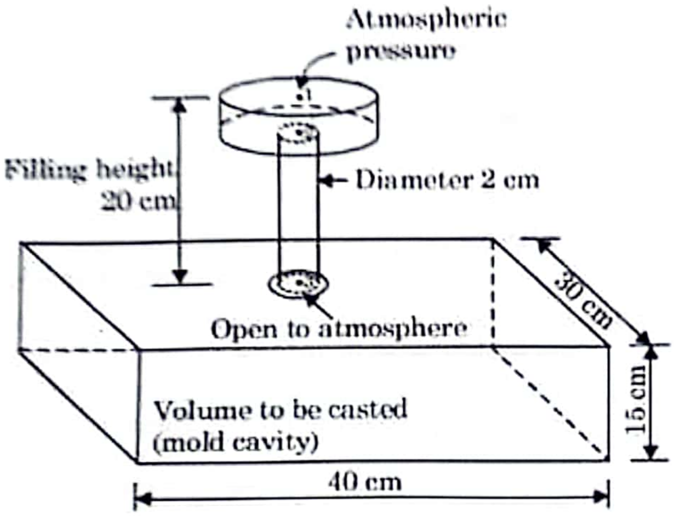
29. In a casting process, a vertical channel through which molten metal flows downward from pouring basin to runner for reaching the mould cavity is called

(a) riser (b) blister

(c) pin hole (d) sprue

[ME 2019 : 1 Mark, Set-1]

30. The figure shows a pouring arrangement for casting of a metal block. Frictional losses are negligible. The acceleration due to gravity is 9.81 m/s^2. The time (in s, round off to two decimal places) to fill up the mold cavity (of size 40 cm × 30 cm × 15 cm) is ____.



[ME 2019 : 2 Marks, Set-2]

31. What is the velocity of the steel at the bottom of the sprue if the sprue height is 30 cm? Assume the frictional and other losses to be 20%.

[PI 1989 : 2 Marks]

32. **Assertion (A):** Converging passage is used for feeding liquid metal into a mould.

Reason (R): Inhalation of air promotes blow holes in casting.

(a) Both (A) and (R) are true and (R) is the correct reason for (A)

(b) Both (A) and (R) are true but (R) is NOT the correct reason for (A)

(c) Both (A) and (R) are false

(d) (A) is false but (R) is true

[PI 1990 : 1 Mark]

33. The optimum pouring time for a casting depends on several factors. One important factor among them is

(a) location of riser

(b) porosity of sand mould

(c) fluidity of casting metal

(d) area of the pouring basin

[PI 1991 : 1 Mark]

34. The primary function of a riser is to

(a) feed molten metal to casting as it solidifies

(b) prevent atmospheric air from contaminating the metal in the mould

(c) allow gases to easily escape from mould cavity

(d) allow molten metal to rise above the mould cavity

[PI 1992 : 1 Mark]

35. Chaplets are placed between mould in order to

(a) promote directional solidification

(b) help alloying the metal

(c) facilitate easy removal of core from casting

(d) prevent core movement due to buoyancy

[PI 1992 : 1 Mark]

36. Application

A. Undercut in components

B. Large bells

C. Mass production of casting by machine moulding

D. Components with irregular parting lines

Type of pattern

1. Cope and drag

2. Follow board

3. Gated

4. Loose piece

5. Sweep

[PI 1992 : 2 Marks]

37. According to Chvorinov's rule, the solidification time of a casting is proportional to (Volume/surface area)^n, where 'n' equals to

(a) 0.5 (b) 1

(c) 2 (d) 4

[PI 1994 : 2 Marks]

38. The height of the down sprue is 175 mm and its CS are at the base is 200 mm^2. The CS area of the horizontal runner is also 200 mm^2. Assuming no losses, indicate the correct choice for the time (second) required to fill a mould cavity of volume 10^6mm^3. (Use g = 10 m/s^2)

(a) 2.67 (b) 8.45

(c) 26.7 (d) 84.50

[PI 2002 : 2 Marks]

39. The permeability of moulding sand was determined using a standard AFS sample by passing 2000 cc of air at a gauge pressure of 10 g/cm^2. If the time taken for the air to escape was 1 min, the permeability number is

(a) 112.4 (b) 100.2

(c) 75.3 (d) 50.1

[PI 2002 : 2 Marks]

40. Proper gating design in metal casting

P. Influences the freezing range of the melt

Q. Compensates the loss of fluidity of the melt

R. Facilitates top feeding of the melt

S. Avoids misrun

(a) P, R (b) Q, S

(c) R, S (d) P, S

[PI 2002 : 2 Marks]

41. A 10 mm thick steel bar is to be horizontally cast with two correctly top risers of adequate feeding capacity. Assuming end effect without chill, what should be the theoretical length of the bar?

(a) 96 mm (b) 132 mm

(c) 192 mm (d) 156 mm

[PI 2002 : 2 Marks]

42. Gating ratio of 1 : 2 : 4 is used to design the gating system for magnesium alloy casting. This gating ratio refers to the cross-section areas of the various gating elements as given below:

1. Down sprue
2. Runner
3. Ingates

The sequence of the above elements in the ratio 1 : 2 : 4 is

(a) 1, 2 and 3 (b) 1, 3 and 2

(c) 2, 3 and 1 (d) 3, 1 and 2

[PI 2003 : 2 Marks]

43. A casting of size 100 mm × 100 mm × 50 mm is required. Assume volume shrinkage of casting as 2.6%. If the height of the riser is 80 mm and riser volume desired is 4 times the shrinkage in casting, what is the appropriate riser diameter in mm?

(a) 14.38 (b) 20.34

(c) 28.76 (d) 57.52

[PI 2003 : 2 Marks]

44. Wood flour is added to core sand to improve

(a) Collapsibility of core

(b) Dry strength of core

(c) Shear strength of core

(d) Tolerance on casting

[PI 2004 : 1 Mark]

45. The shape factor for a casting in the form of an annular cylinder of outside diameter 30 cm, inside diameter 20 cm and height 30 cm (correction factor k = 1.0) will be

(a) 21.77 (b) 6.28

(c) 9.42 (d) 12.28

[PI 2005 : 2 Marks]

46. A cast Steel slab of dimension 30 × 20 × 5 cm is poured horizontally using a side riser. The riser is cylindrical in shape with diameter and height, both equal to D. The freezing ratio of the mould is

(a) 8D/75 (b) 4D/75

(c) 75/8D (d) 75/4D

[PI 2005 : 2 Marks]

47. Cold shut is a defect in casting due to

(a) sand sliding from the cope surface

(b) internal voids or surface depression due to excessive gas trapped

(c) discontinuity resulting from hundred contraction

(d) two streams of material that are too cold to fuse properly

[PI 2006 : 1 Mark]

48. The volume flow rate (in mm^3/s) is

(a) 0.8×10^5 (b) 1.1×10^5

(c) 1.7×10^5 (d) 2.3×10^5

[PI 2007 : 2 Marks

49. The mould filling time is seconds is

(a) 2.8 (b) 5.78

(c) 7.54 (d) 8.41

[PI 2007 : 2 Marks

13. In a sand casting operation, the total liquid head is maintained constant such that it is equal to the mould height. The time taken to fill the mould with a top gate is t_A. If the same mould is filled with a bottom gate, then the time taken is t_B. Ignore the time required to fill the runner and frictional effects, Assume Atmospheric pressure at the top molten metal surfaces. The relation between t_A and t_B is

(a) $t_B = \sqrt{2t_A}$ (b) $t_B = 2t_A$

(c) $t_B = \dfrac{t_A}{\sqrt{2}}$ (d) $t_B = 2\sqrt{2}t_A$

[ME 2006 : 2 Marks]

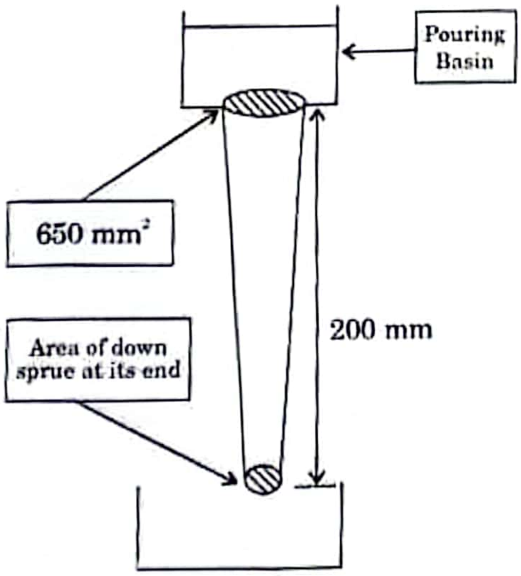
14. Volume of a cube of side l and volume of a sphere of radius r are equal. Both the cube and the sphere are solid and of same material. They are being cast. The ratio of the solidification time of the cube to the same of sphere is

(a) $\left(\dfrac{2\pi}{6}\right)^3\left(\dfrac{r}{l}\right)^6$ (b) $\left(\dfrac{4\pi}{6}\right)\left(\dfrac{r}{l}\right)^2$

(c) $\left(\dfrac{4\pi}{6}\right)^2\left(\dfrac{r}{l}\right)^3$ (d) $\left(\dfrac{4\pi}{6}\right)^2\left(\dfrac{r}{l}\right)^4$

[ME 2007 : 1 Mark]

15. A 200 mm long down sprue has an area of cross-section of 650 mm² where the pouring basin meets the down sprue (i.e. at the beginning of the down sprue). A constant head of molten metal is maintained by the pouring basin. The molten metal flow rate is 6.5×10^5 mm³/s. Considering the end of down sprue to be open to atmosphere and acceleration due to gravity of 10^4 mm/s², the area of the down sprue in mm² at its end (avoiding aspiration effect) should be



(a) 650.0 (c) 290.7

(b) 350.0 (d) 190.0

[ME 2007 : 2 Marks]

16. Match the items in Column I and Column II.

Column I	Column II
P. Metallic Chills	1. Support for the core
Q. Metallic Chaplets	2. Reservoir of the molten metal
R. Riser	3. Control cooling of critical sections
S. Exothermic Padding	4. Progressive solidification

(a) P–1, Q–3, R–2, S–4

(b) P–1, Q–4, R–2, S–3

(c) P–3, Q–4, R–2, S–1

(d) P–4, Q–1, R–2, S–3

[ME 2009 : 2 Marks]

17. In a gating system, the ratio 1:2:4 represents

(a) sprue base area : runner area : ingate area

(b) pouring basin area: ingate area: runner area

(c) sprue base area : ingate area : casting area

(d) runner area : ingate area : casting area

[ME 2010 : 1 Mark]

18. A cubic casting of 50 mm side undergoes volumetric solidification shrinkage and volumetric solid contraction of 4% and 6% respectively. No riser is used. Assume uniform cooling in all directions. The side of the cube after solidification and contraction is

(a) 48.32 mm (b) 49.90 mm

(c) 49.94 mm (d) 49.96 mm

[ME 2011 : 2 Marks]

19. A cube shaped casting solidifies in 5 min. The solidification time in min for a cube of the same material, which is 8 times heavier than the original casting, will be

(a) 10 (b) 20

(c) 24 (d) 40

[ME 2013 : 1 Mark]

Equation (2.7) gives the velocity of a jet discharging against a static head h, making the effective head as $(h_t - h)$. Now, for the instant shown, let the metal level in the mould move up through a height dh in a time interval dt; A_m and A_g are the cross-sectional areas of the mould and the gate, respectively. Then,

$$A_m \, dh = A_g v_g \, dt. \tag{2.8}$$

Using equations (2.7) and (2.8), we get

$$\frac{1}{\sqrt{2g}} \frac{dh}{\sqrt{h_t - h}} = \frac{A_g}{A_m} dt. \tag{2.9}$$

At $t = 0$, $h = 0$ and at $t = t_f$ (filling time), $h = h_m$. Integrating equation (2.9) between these limits, we have

$$\frac{1}{\sqrt{2g}} \int_0^{h_m} \frac{dh}{\sqrt{h_t - h}} = \frac{A_g}{A_m} \int_0^{t_f} dt$$

or

$$t_f = \frac{A_m}{A_g} \frac{1}{\sqrt{2g}} 2(\sqrt{h_t} - \sqrt{h_t - h_m}). \tag{2.10}$$

If a riser (reservoir to take care of the shrinkage from the pouring temperature) is used, then the pouring time t_f should also include the time needed to fill up the riser. Normally, open risers are filled up to the level of the pouring sprue; thus, the time taken to fill up the riser is calculated with A_m replaced by A_r (riser cross-section) and h_m by h_t in equation (2.10).

EXAMPLE 2.1 Two gating designs for a mould of 50 cm × 25 cm × 15 cm are shown in Fig. 2.7. The cross-sectional area of the gate is 5 cm². Determine the filling time for both the designs.

SOLUTION Figure 2.7a. Since $h_t = 15$ cm, from equation (2.3), we have

$$v_3 = \sqrt{2 \times 981 \times 15} \text{ cm/sec} = 171.6 \text{ cm/sec}.$$

The volume of the mould is $V = 50 \times 25 \times 15$ cm³ and the cross-sectional area of the gate is $A_g = 5$ cm². So, from equation (2.4), we get

$$t_f = \frac{50 \times 25 \times 15}{5 \times 171.6} \text{ sec} = 21.86 \text{ sec}.$$

Figure 2.7b. Here, $h_t = 15$ cm, $h_m = 15$ cm, $A_m = 50 \times 25$ cm², and $A_g = 5$ cm². Using equation (2.10), we have

$$t_f = \frac{50 \times 25}{5} \frac{\sqrt{2}}{\sqrt{981}} \sqrt{15} \text{ sec} = 43.71 \text{ sec}.$$

It should be noted that in Fig. 2.7b the time taken is double of that in Fig. 2.7a. We can easily verify that this will always be so if $h_m = h_t$.

say, 1 and 2, in that order in the direction of flow, E_{f_1} should be added to the energy of station 2.

For a smooth conduit, the value of f is given by the equations

$$f = 16/Re \quad \text{for laminar flow} \qquad (Re < 2000), \tag{2.17}$$

$$\frac{1}{\sqrt{f}} = 4\log_{10}(Re\sqrt{f}) - 0.4 \quad \text{for turbulent flow} \qquad (Re > 2000), \tag{2.18}$$

where Re is Reynolds number. For the range $2100 < Re < 10^5$, equation (2.18) can be simplified to the form

$$f = 0.0791(Re)^{-1/4}. \tag{2.19}$$

Frictional losses also occur due to a gradual change in the flow direction, e.g., in a 90° bend and other similar fittings. In such a situation, the associated frictional losses are accounted for by using equation (2.16) with an equivalent (L/D) factor for the bend.

The frictional loss (per unit mass) associated with a sudden enlargement or contraction of a flow area is expressed as

$$E_{f_2} = \tfrac{1}{2}e_f\bar{v}^2, \tag{2.20}$$

where $\bar{v}$ is the average velocity of the fluid in the smaller cross-section, and e_f is the friction loss factor and it depends on the ratio of the flow area and Reynolds number of the flow. For a laminar flow, the length and diameter of the smaller flow cross-section have also to be taken into account. The value of e_f depends on whether the flow area is enlarging or contracting in the direction of flow. The values of e_f for a sharp change in the flow cross-section are shown in Figs. 2.11a and 2.11b for sudden expansion and contraction, respectively. In these figures, the Reynolds numbers of the flow have been calculated on the basis of the average flow velocity in the smaller cross-sections, and l and d are the length and diameter, respectively, of the smaller flow cross-sections. The values of e_f for some other types of changes in flow geometry are listed in Fig. 2.12; here, e_{f_s} refers to the values corresponding to a sharp change of geometry with identical initial and final dimensions (i.e., the values of e_{f_s} can be obtained from Figs. 2.11a and 2.11b). Now, going back to Fig. 2.6a, let us modify the analysis presented in Section 2.4 in the light of the effects of friction and velocity distribution.

Using the integrated energy balance equation between points 1 and 3, after accounting for the loss due to a sudden contraction at 2, we get

$$\frac{p_1}{\rho_m} + 0 + gh_t = \frac{p_3}{\rho_m} + \frac{\bar{v}_3^2}{2\beta} + E_{f_1} + E_{f_2}, \tag{2.21}$$

where $\bar{v}_3$ is the average fluid velocity in the sprue. With $p_1 = p_3$ and using equations (2.16) and (2.20), we get

$$2gh_t = \bar{v}_3^2\left(\frac{1}{\beta} + e_f + 4f\frac{l}{d}\right), \tag{2.22}$$

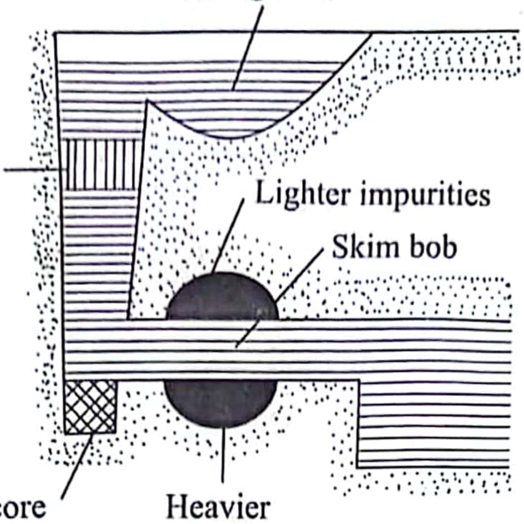


Fig. 2.10 Gating design to prevent impurities.

2.4.2 EFFECTS OF FRICTION AND VELOCITY DISTRIBUTION

In Sections 2.4 and 2.4.1, we assumed that the velocity of a liquid metal in t sprue and the gate is uniform across the cross-section. In fact, the velocity a fluid in contact with any solid surface is zero and is maximum at the axis the conduit. The velocity distribution within the conduit depends on the sha of the conduit and the nature of the flow (i.e., turbulent or laminar). Further, our discussion so far, we have also assumed no frictional losses. In real fluid the frictional losses are always present, especially when there is a sudd contraction in or an enlargement of the flow cross-sections. In the discussic that follows, we shall, in the light of these two factors, i.e., velocity distributic and friction, modify the equations we have already developed.

The nonuniform velocity distribution can be accounted for by modifying t kinetic energy term in the integrated energy balance equation by replacing t (velocity)^2 term by $\bar{v}^2/\beta$, where $\bar{v}$ is the average velocity and β is a constan For a circular conduit, the value of β is equal to 0.5 for laminar flow an approximately equal to 1 for turbulent flow.

The energy loss due to friction in a circular conduit (on the basis of per un mass) is given by

$$E_{f_1} = 4f\frac{L}{D}\frac{\bar{v}^2}{2}, \qquad (2.16$$

where

$\bar{v}$ = average velocity,

D, L = diameter and length, respectively, of the conduit, and

f = friction factor.

he value of f depends on the roughness of the conduit and the nature of the ow. Thus, while using the integrated energy equation between two points,

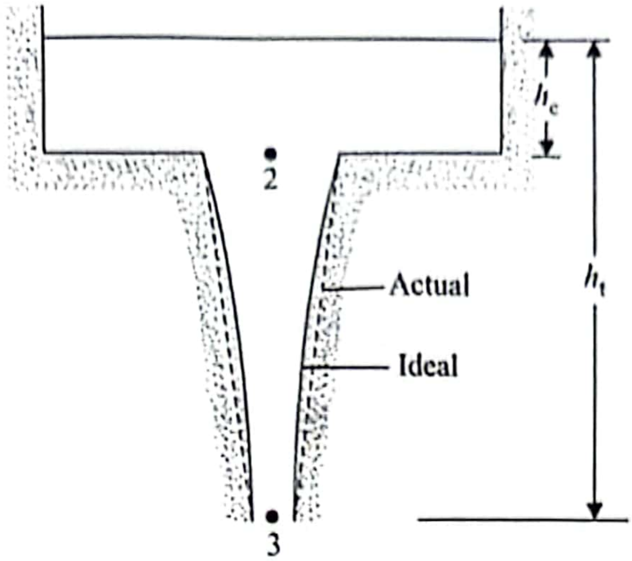
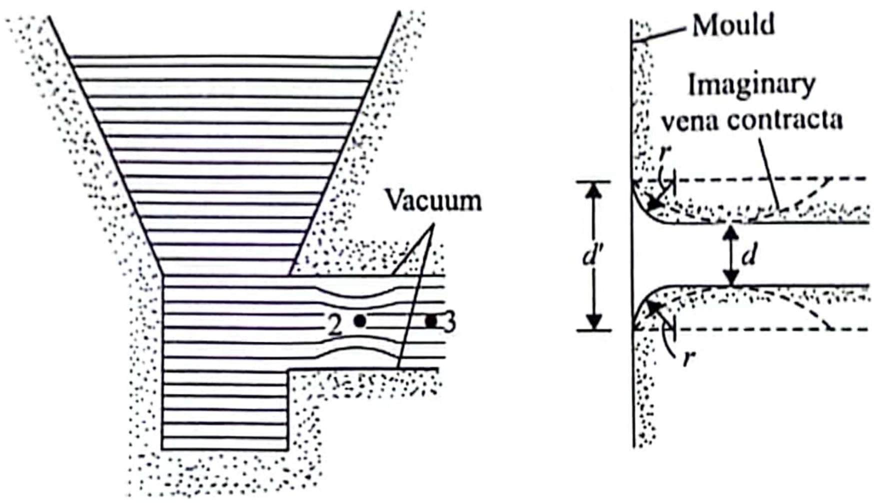


Fig. 2.8 Ideal and actual shapes of sprue.



(a) Mechanism of vacuum generation

(b) Outlet dimensions to prevent vacuum generation

Fig. 2.9 Principle of avoiding vacuum generation.

The common items employed in a gating design to prevent impurities in the casting are as follows (see also Fig. 2.10).

(i) *Pouring basin* This reduces the eroding force of the liquid metal stream coming directly from the furnace. A constant pouring head can also be maintained by using a pouring basin.

(ii) *Strainer* A ceramic strainer in the sprue removes dross.

(iii) *Splash core* A ceramic splash core placed at the end of the sprue also reduces the eroding force of the liquid metal stream.

(iv) *Skim bob* It is a trap placed in a horizontal gate to prevent heavier and lighter impurities from entering the mould.

Let, in the limiting case, p_2 be equal to zero, when, from equation (2.11),

$$\frac{v_3^2}{2} = gh_2 + \frac{v_2^2}{2}. \qquad (2.12)$$

From the principle of continuity of flow, $A_2v_2 = A_3v_3$, where A is the cros sectional area. Thus,

$$v_2 = \frac{A_3}{A_2}v_3 = Rv_3, \qquad (2.13)$$

where $R = A_3/A_2$. Using equations (2.12) and (2.13), we obtain

$$\frac{v_3^2}{2g} = h_2 + \frac{R^2v_3^2}{2g}$$

or

$$R^2 = 1 - \frac{2gh_2}{v_3^2}. \qquad (2.14)$$

Again, $v_3^2 = 2gh_t$ (applying Bernoulli's equation between points 1 and 3, wi $p_1 = p_3 = 0$ and $v_1 = 0$). Substituting this in equation (2.14), we have

$$R^2 = 1 - \frac{h_2}{h_t} = \frac{h_c}{h_t}$$

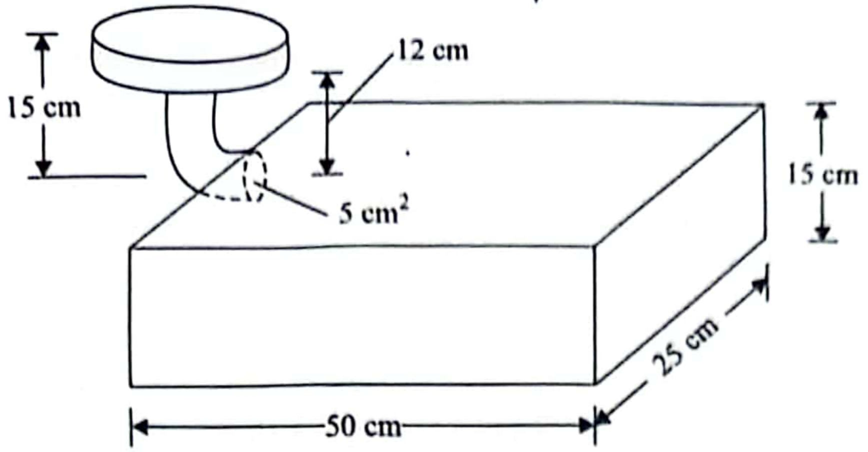
or

$$R = \frac{A_3}{A_2} = \sqrt{\frac{h_c}{h_t}}. \qquad (2.15)$$

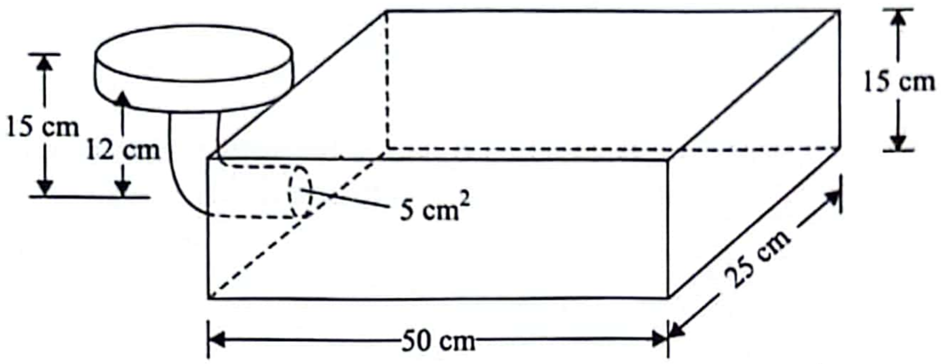
[This can easily be seen to be the shape of a freely falling stream whe $v_2 = \sqrt{2gh_c}$ and $v_3 = \sqrt{2gh_t}$.] Thus, ideally, the sprue profile should be as show by the solid lines in Fig. 2.8 when the pressure throughout the stream is ju atmospheric. However, a straight tapered sprue (shown by the dashed lines) i safer (pressure everywhere, except at points 2 and 3, is above atmospheric) an easier to construct. The sprue design in Fig. 2.6b is better than that in Fig. 2.6a

Another situation where aspiration effect comes into the picture is associate with a sudden change in the flow direction. As shown in Fig. 2.9a, the liqui metal stream contracts around a sharp corner due to the momentum effect. I vertical gating, this has got nothing to do with acceleration due to gravity. Th constricted region shown at station 2 in Fig. 2.9a is known as *vena contracta* To avoid the creation of vacuum around station 2, the mould is made to fit th vena contracta, as done in Fig. 2.9b. In other words, a sharp change in the flov direction is avoided. If the runner diameter is d and the diameter at the entranc is d', then, normally, d'/d is maintained at a value approximately equal to 1.3[1] This means $r \approx 0.15d$.

[1]Flinn, R.A., Fundamentals of Metal Casting, Addison-Wesley, Reading, Massachusetts 1963.



(a) Top gating



(b) Bottom gating

Fig. 2.7 Top and bottom gating designs.

2.4.1 ASPIRATION EFFECT

For a mould made of a permeable material (e.g., sand), care should be taken to ensure that the pressure anywhere in the liquid metal stream does not fall below the atmospheric pressure. Otherwise, the gases originating from baking of the organic compounds in the mould will enter the molten metal stream, producing porous castings. This is known as the aspiration effect.

Referring to Fig. 2.6a and applying Bernoulli's equation between points 2 and 3, we obtain

$$gh_2 + \frac{p_2}{\rho_m} + \frac{v_2^2}{2} = \frac{p_3}{\rho_m} + \frac{v_3^2}{2}, \tag{2.11}$$

where p and v refer to the pressure and velocity, respectively, of the liquid metal at stations 2 and 3. If the pressure at point 3 is atmospheric, i.e., $p_3 = 0$, then $p_2 = -\rho_m g h_2$ as $v_2 = v_3$. Hence, the design in Fig. 2.6a is not acceptable. To avoid negative pressure at point 2 (to ensure positive pressure anywhere in the liquid column), the sprue should be tapered, the ideal shape of which can be determined as follows.